The Chinese University of Hong Kong, Shenzhen

Ordinary Differential Equation Notes

Author: Shuo Guo

July 25, 2026

Contents

Chapter 1

First Order Differential Equations

For the first order differential equations, we would consider to solve

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y)$$

Consider a simpler case:

$$\frac{\mathrm{d}}{\mathrm{d}} = g(t)$$

One with elementary calculus knowledge can solve it:

$$y(t) = \int g(t)dt + c$$

However, for general first order equation, it's very hard to solve.
We would first consider some simpler cases.

1 First order linear differential equations

Consider f is consisted by linear equations.

Definition 1.1.1 (First-oder linear differential equation)**.**

The general first-order linear differential equation is

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = b(t) \tag{1}$$

Needless to say, $a(t)$, $b(t)$ are assumed to be continuous functions of time. To slove general first order linear equation(1), we could consider a simpler case:

Definition 1.1.2 (Homogeneous first order linear equation)**.**
The equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = 0 \tag{2}$$

is called homogeneous first-order linear differential equation

1.1 Homogeneous first order linear differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = 0.$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t)y.$$
$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t).$$

Integrating both sides with respect to t gives

$$\int \frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t}\, dt = -\int a(t)\, dt.$$

By the chain rule, $\frac{1}{y}\frac{dy}{dt} = \frac{d}{dt}\ln|y|$, hence

$$\ln|y(t)| = -\int a(t)\, dt + C,$$

where C is an arbitrary constant of integration. Exponentiating both sides yields

$$|y(t)| = e^{C} e^{-\int a(t)\, dt}.$$

Since $e^{C} > 0$, we may absorb the sign into a new constant $c = \pm e^{C}$, obtaining the general solution

$$\boxed{y(t) = c\, e^{-\int a(t)\, dt}} \tag{3}$$

where $c \in \mathbb{R}$ is an arbitrary constant. Note that the zero solution corresponds to $c = 0$.

In most cases, we are not interest in all solutions of (3). We could add more assmup-tion as

$$y(t_0) = y_0 \tag{4}$$

Then we integral (3) between t_0 and t.
Recalling $\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t)$, we could write:

$$\int_{t_0}^{t} \frac{\mathrm{d}}{\mathrm{d}s}\ln|y(s)|ds = \int_{t_0}^{t} -a(s)ds$$

So

$$\ln|y(t)| - \ln|y(t_0)| = \ln|\frac{y(t)}{y(t_0)}| = -\int_{t_0}^{t} a(s)ds$$

Thus,

$$|\frac{y(t)}{y(t_0}| = \exp(-\int_{t_0}^{t} a(s)ds)$$

The problem now becomes +1 or -1. By taking $t = t_0$, we can find it's +1.
So

$$y(t) = y(t_0)\exp(\int_{t_0}^{t} a(s)ds) \tag{5}$$

1.2 General first order linear differential equation

Consider the problem:

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = b(t) \tag{6}$$

We want to write it in a form:

$$\frac{\mathrm{d}'something'}{\mathrm{d}t} = c(t)$$

and then we can solve it. Multiply both sides by a continuous function $\mu(t)$ [1]
So

$$\mu(t)\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)\mu(t)y = \mu(t)b(t)$$

We consider another formula: $\frac{\mathrm{d}}{\mathrm{d}t}\mu(t)y = \frac{\mathrm{d}y}{\mathrm{d}t}\mu(t) + y\frac{\mathrm{d}\mu(t)}{\mathrm{d}t}$.
So we try to choose proper μ s.t. the left of the equation can be written as $\frac{\mathrm{d}'something'}{\mathrm{d}t}$.
We choose

$$\mu(t) = \exp(\int a(t)dt) \tag{7}$$

And

$$\frac{\mathrm{d}}{\mathrm{d}t}\mu(t)y = \mu(t)b(t)$$

$$\mu(t)y = \int \mu(t)b(t)dt + c$$

$$y = \frac{1}{\mu(t)}(\int \mu(t)b(t)dt + c), \quad \text{with } \mu(t) = \exp(\int a(t)dt) \tag{8}$$

We could consider $y(t_0) = y_0$ So:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mu(t)y = \mu(t)b(t)$$

$$\Downarrow$$

$$\mu(t)y - \mu(t_0)y_0 = \int_{t_0}^{t} \mu(s)b(s)ds$$

Thus,

$$y = \frac{1}{\mu(t)}\Big(\mu(t_0)y_0 + \int_{t_0}^{t} \mu(s)b(s)dt\Big) \tag{9}$$

[1]Here any continuous μ could satisfies the condition.

2 Separable equations

Recall how we deal with linear differential equation:

$$\frac{1}{y(t)}\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t)$$

$$\Downarrow$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\ln|g(t)| = -a(t)$$

So for a more general ODE with the form:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{g(t)}{f(y)} \tag{10}$$

We could write it as

$$\frac{\mathrm{d}}{\mathrm{d}t}F(y(t)) = g(t), \quad \text{with } F(y) = \int f(y)dy$$

So the answer is:

$$F(y(t)) = \int g(t)dt + c \tag{11}$$

We could also add an assumption $y(t_0) = y_0$
So

$$F(y(t)) - F(y(t_0)) = \int_{t_0}^{t} g(s)ds$$

$$\int_{y_0}^{y} f(r)dr = \int_{t_0}^{t} g(s)ds$$

3 Population models

Let $p(t)$ denotes the population of a given species at time t, $r(t, p)$ denotes the difference between its birth rate and death rate.
For the simplest, we think that r will not change with t and p, which means it's a constant. The simplest model:

$$\frac{\mathrm{d}p(t)}{\mathrm{d}t} = ap(t), \quad a \text{ is a constant}$$

Also, assume $p(t_0) = p_0$
We can easily find the solution

$$p(t) = p_0 \exp(a(t - t_0)) \tag{12}$$

This result works very well if the population is small. However, if the population grows larger and larger, it falls. So we can let

$$\frac{\mathrm{d}r}{\mathrm{d}t} = ar - br^2 \tag{13}$$

It's also a sparable problem, though the integer part is a bit complex (but absolutly solvable!)
The answer is:

$$p(t) = \frac{ap_0 e^{a(t-t_0)}}{a - bp_0 + bp_0 e^{a(t-t_0)}} = \frac{ap_0}{bp_0 + (a - bp_0)e^{-a(t-t_0)}}.$$

4 Exact equations and the range we can solve

Consider all differential equation we've solved, we can put them in this form:

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi(t,y) = 0 \tag{14}$$

Example 1.4.1 (ϕ for first order linear differential equations)**.**
Thinking about all sparable differential equation (it would also contains all first order linear differential equations) we've solved before:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = g(t)h(y(t))$$

We can all write this equation in this form:

$$\phi(t,y) = \int^y \frac{1}{h(s)}ds - \int^t g(r)dr$$

More generally, we can solve all differential equations which can be written in this form:

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi(t,y(t)) = 0 \tag{15}$$

Remark 4.1. Actually, 'we can solve' doesn't mean that we can get explict solution. It just becoma a equation without differential term. But with the aid of computer, we can solve its numer with any desired accuracy.

Consider:

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi(t,y) = \frac{\partial\phi}{\partial t} + \frac{\partial\phi}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t} \tag{16}$$

One natural question is that for all $M(t,y), N(t,y)$, does there exist ϕ s.t.

$$M = \frac{\partial\phi}{\partial t} \quad N = \frac{\partial\phi}{\partial y} \tag{17}$$

The answer is **NO**.

Theorem 1.4.2. *Let $M(t,y)$ and $N(t,y)$ be continuous and have continuous partial derivatives with respect to t and y in the rectangle R consisting of those points (t,y) with $a < t < b$ and $c < y < d$. There exists a function $\phi(t,y)$ such that*

$$M(t,y) = \frac{\partial\phi}{\partial t} \quad \textit{and} \quad N(t,y) = \frac{\partial\phi}{\partial y}$$

if, and only if,

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}$$

in R.

The entire proof will be in 4.3.

Definition 1.4.1 (Exact equations)**.**
The differential equation

$$M(t,y)+N(t,y)\frac{dy}{dt}=0$$

is said to be exact *if*

$$\frac{\partial M}{\partial y}=\frac{\partial N}{\partial t}.$$

For exact equations, we can write:

$$\phi(t,y)=\int M(t,y)+\int\Big(N(t,y)-\int\frac{\partial M(t,y)}{\partial y}dt\Big)dy \tag{18}$$

Remark 4.3. It's not necessary that $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial t}$ in a rectangle R as theorem 1.4.2 states. It can be any region without 'hole' inside.

Remark 4.4. For differential equation $dy/dt - f(t,y)$, it can be written in the form $M(t,y)+N(t,y)(dy/dt)=0$ by setting $M(t,y)=-f(t,y)$ and $N(t,y)=1$.

Now we've solved Exact equations. Suppose

$$M(t,y)+N(t,y)\frac{\mathrm{d}y}{\mathrm{d}t}=0 \tag{19}$$

is not exact. Could we solve it?
A natural idea is to multiply both terms with a function $\mu(t,y)$:

$$M(t,y)\mu(t,y)+\mu(t,y)N(t,y)\frac{\mathrm{d}y}{\mathrm{d}t}=0 \tag{20}$$

And if μ is well-chosen, it could become an exact equation, and we can have

$$\frac{\partial}{\partial y}\Big(M(t,y)\mu(t,y)\Big)=\frac{\partial}{\partial t}\Big(N(t,y)\mu(t,y)\Big) \tag{21}$$

Definition 1.4.2 (Integrating Factor)**.**
Any function $\mu(t,y)$ satisfying (21) is called integrating factor for the equation (19)

Then we try to find μ.

$$\frac{\partial}{\partial y}\Big(M(t,y)\mu(t,y)\Big)=\frac{\partial}{\partial t}\Big(N(t,y)\mu(t,y)\Big)$$

$$\Downarrow$$

$$M\frac{\mathrm{d}}{\mathrm{d}y}\mu+\mu\frac{\mathrm{d}}{\mathrm{d}y}M=N\frac{\mathrm{d}\mu}{\mathrm{d}t}+\mu\frac{\mathrm{d}}{\mathrm{d}t}N$$

$$\Downarrow$$

$$-N\frac{\mathrm{d}\mu}{\mathrm{d}t} + M\frac{\mathrm{d}\mu}{\mathrm{d}y} = (\frac{\mathrm{d}N}{\mathrm{d}t} - \frac{\mathrm{d}M}{\mathrm{d}y})\mu$$

Solving a general $\mu(t, y)$ is almost as difficult as solving the ODE equation.
So if μ could only depend on t or y, we could be able to solve it. **It's just an assmuption. We cannot say that there must exists $\mu = \mu(t)$ or $\mu = \mu(y)$. Just for some special cases, we calculate some μ below and then we find if we are lucky enough to obtain $\mu = \mu(t)$ or $\mu = \mu(y)$.**

4.1 $\mu = \mu(t)$

If μ depends only on t, then $\partial\mu/\partial y = 0$. Therefore,

$$-N\frac{d\mu}{dt} = \left(\frac{\partial N}{\partial t} - \frac{\partial M}{\partial y}\right)\mu,$$

and hence

$$\frac{1}{\mu}\frac{d\mu}{dt} = \frac{\dfrac{\partial M}{\partial y} - \dfrac{\partial N}{\partial t}}{N}. \tag{22}$$

Since the left-hand side depends only on t, the expression on the right-hand side must also be a function of t alone. If

$$\frac{\dfrac{\partial M}{\partial y} - \dfrac{\partial N}{\partial t}}{N} = f(t),$$

then integrating (22) gives

$$\boxed{\mu(t) = C\exp\left(\int f(t)\,dt\right)}, \qquad C \neq 0. \tag{23}$$

4.2 $\mu = \mu(y)$

If μ depends only on y, then $\partial\mu/\partial t = 0$. Therefore,

$$M\frac{d\mu}{dy} = \left(\frac{\partial N}{\partial t} - \frac{\partial M}{\partial y}\right)\mu,$$

and hence

$$\frac{1}{\mu}\frac{d\mu}{dy} = \frac{\dfrac{\partial N}{\partial t} - \dfrac{\partial M}{\partial y}}{M}. \tag{24}$$

Since the left-hand side depends only on y, the expression on the right-hand side must also be a function of y alone. If

$$\frac{\dfrac{\partial N}{\partial t} - \dfrac{\partial M}{\partial y}}{M} = g(y),$$

then integrating (24) gives

$$\mu(y) = C \exp\left(\int g(y)\,dy\right), \qquad C \neq 0. \tag{25}$$

4.3 Proof of theorem1.4.2

Proof. Fix $t_0 \in (a,b)$. First, suppose that ϕ satisfies

$$\frac{\partial \phi}{\partial t} = M(t,y).$$

Integrating with respect to t, while regarding y as a constant, gives

$$\phi(t,y) = \int_{t_0}^{t} M(s,y)\,ds + h(y), \text{ where } h(y) \text{ is an arbitrary function depending only on } y. \tag{26}$$

Differentiating (26) with respect to y, we obtain

$$\frac{\partial \phi}{\partial y} = \int_{t_0}^{t} \frac{\partial M}{\partial y}(s,y)\,ds + \frac{dh}{dy}(y).$$

We also require $\partial\phi/\partial y = N(t,y)$. Therefore,

$$\frac{dh}{dy}(y) = N(t,y) - \int_{t_0}^{t} \frac{\partial M}{\partial y}(s,y)\,ds. \tag{27}$$

The left-hand side of (27) depends only on y, so the right-hand side must also be independent of t. Differentiating the right-hand side with respect to t, we find

$$\frac{\partial}{\partial t}\left(N(t,y) - \int_{t_0}^{t} \frac{\partial M}{\partial y}(s,y)\,ds\right) = \frac{\partial N}{\partial t}(t,y) - \frac{\partial M}{\partial y}(t,y) = 0$$

Thus, if such a function ϕ exists, then necessarily

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}.$$

Conversely, if this equality holds throughout R, then the right-hand side of (27) has zero derivative with respect to t, and hence depends only on y. We can therefore integrate it with respect to y to obtain a function $h(y)$. Substituting this h into (26) gives a function satisfying

$$\frac{\partial \phi}{\partial t} = M(t,y) \quad \text{and} \quad \frac{\partial \phi}{\partial y} = N(t,y).$$

Consequently, such a function ϕ exists if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}.$$

$\square$

5 The existence-uniqueness theorem: Picard iteration

Many first order differential equations (actually, most) we cannot get the explict solution. However, can we prove that there exists a solution? If so, could we prove that it's the only sulution, or it's the only solution under some conditions?

Consider

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y), \quad y(t_0) = y_0 \tag{28}$$

So

$$y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))ds \quad \text{as } y(t) \text{ is continuous and } y(t_0) = y_0 \tag{29}$$

Define an operator[2]

$$\mathcal{L}[y](t) = y_0 + \int_{t_0}^{t} f(s, y(s))ds$$

$$\mathcal{L} : C([0, T]) \to C([0, T])$$

For equation

$$y(t) = \mathcal{L}[y](t)$$

we call it *fixed point* of $\mathcal{L}$.
So the problem: Finding the solution of ODE $\to$ Finding the fixed point of $\mathcal{L}$. We'll use a sequence of $\{y_n\}$ to approximate.

In the following subsubsections, we will learn how to construct a sequence of $\{y_n\}$ in 5.1. We'll talk about whether the sequence will converge in 5.2 (But we don't care where it converges). We'll prove that the convergence result is the solution of (29) in 5.3. Last, we'll prove the uniqueness of the solution in 5.4

5.1 Construction of Picard iterates

To solve

$$y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))ds \quad \text{as } y(t) \text{ is continuous and } y(t_0) = y_0 \tag{30}$$

we create a sequance of $\{y(t)\}$ to approximate the answer. We start with the constant function

$$y_0(t) = y_0.$$

Substituting this initial guess into the right-hand side of the integral equation gives the first Picard iterate:

$$y_1(t) = \mathcal{L}[y_0](t) = y_0 + \int_{t_0}^{t} f(s, y_0(s))\, ds = y_0 + \int_{t_0}^{t} f(s, y_0)\, ds.$$

[2] Actually I still don't get fully understand why we do so complex here. It may have some connection with the space? Anyway, see it, accept it, and see what's going on later.

If $y_1(t) = y_0(t)$, then y_0 is already a fixed point of $\mathcal{L}$, so it is a solution of the initial value problem. If not, we substitute y_1 into $\mathcal{L}$ again and obtain

$$y_2(t) = \mathcal{L}[y_1](t) = y_0 + \int_{t_0}^{t} f(s, y_1(s))\, ds.$$

We continue this process. At each step, if

$$y_{n+1}(t) = y_n(t),$$

then y_n is a fixed point of $\mathcal{L}$ and hence a solution. Otherwise, we use y_{n+1} to construct the next iterate. In general, the Picard iterates are defined by

$$\boxed{y_0(t) = y_0, \qquad y_{n+1}(t) = y_0 + \int_{t_0}^{t} f(s, y_n(s))\, ds, \quad n \geq 0.} \tag{31}$$

Notice that every iterate satisfies $y_n(t_0) = y_0$.
Then we'll show the convergence of Picard iterates.

5.2 Convergence of Picard iterates

Lemma 5.1. *Choose any two positive numbers a and b, and let R be the rectangle:*

$$t_0 \leq t \leq t_0 + a, \quad |y - y_0| \leq b.$$

Compute

$$M = \max_{(t,y)\in R} |f(t, y)|, \quad \text{and set } \alpha = \min\left(a, \frac{b}{M}\right).$$

Then,

$$|y_n(t) - y_0| \leq M(t - t_0)$$

for $t_0 \leq t \leq t_0 + \alpha$.

Proof. For $n = 0$,

$$|y_0(t) - y_0| = 0 \leq M(t - t_0).$$

Suppose the result is true for $n = j$. Since $t - t_0 \leq \alpha$,

$$|y_j(t) - y_0| \leq M(t - t_0) \leq M\alpha \leq b.$$

Thus, $(t, y_j(t)) \in R$, so $|f(t, y_j(t))| \leq M$. Therefore,

$$\begin{aligned} |y_{j+1}(t) - y_0| &= \left| \int_{t_0}^{t} f(s, y_j(s))\, ds \right| \\ &\leq \int_{t_0}^{t} |f(s, y_j(s))|\, ds \\ &\leq M(t - t_0). \end{aligned}$$

Hence the result follows for every $n \geq 0$ by induction. □

We now show the convergence of $\{y_n\}$. First, we write y_n in this form:

$$y_n(t) = y_0(t) + [y_1(t) - y_0(t)] + \cdots + [y_n(t) - y_{n-1}(t)]$$

So the sequence $y_n(t)$ converges iff the infinite series converges:

$$[y_1(t) - y_0(t)] + [y_2(t) - y_1(t)] + \cdots + [y_n(t) - y_{n-1}(t)] + \cdots$$

Then, we try to prove a stronger result:

$$\sum_{n=1}^{\infty} |y_n(t) - y_{n-1}(t)| < \infty \Rightarrow \sum_{n=1}^{\infty} \big[y_n(t) - y_{n-1}(t)\big] \text{ converges.} \tag{32}$$

This is not obvious (at least not obvious for me). The result used in (32) will be proved in 5.5.

$$L = \max_{(t,y)\in R} \left| \frac{\partial f}{\partial y}(t, y) \right|.$$

By the mean value theorem, for some $\xi_n(s)$ between $y_n(s)$ and $y_{n-1}(s)$,

$$\begin{aligned}
|f(s, y_n(s)) - f(s, y_{n-1}(s))| &= \left| \frac{\partial f}{\partial y}(s, \xi_n(s)) \right| |y_n(s) - y_{n-1}(s)| \\
&\le L|y_n(s) - y_{n-1}(s)|.
\end{aligned}$$

where

$$L = \max_{(t,y)\in R} \left| \frac{\partial f}{\partial y}(t, y) \right|.$$

First,

$$\begin{aligned}
|y_1(t) - y_0(t)| &= \left| \int_{t_0}^{t} f(s, y_0)\, ds \right| \\
&\le M(t - t_0).
\end{aligned}$$

Then

$$\begin{aligned}
|y_2(t) - y_1(t)| &= \left| \int_{t_0}^{t} \big[f(s, y_1(s)) - f(s, y_0(s))\big]\, ds \right| \\
&\le L \int_{t_0}^{t} |y_1(s) - y_0(s)|\, ds \\
&\le ML \int_{t_0}^{t} (s - t_0)\, ds \\
&= \frac{ML(t - t_0)^2}{2!},
\end{aligned}$$

and

$$\begin{aligned}|y_3(t) - y_2(t)| &\le L \int_{t_0}^{t} |y_2(s) - y_1(s)|\, ds \\ &\le \frac{ML^2(t-t_0)^3}{3!}.\end{aligned}$$

Continuing in the same way,

$$\boxed{|y_n(t) - y_{n-1}(t)| \le \frac{ML^{n-1}(t-t_0)^n}{n!}, \qquad n \ge 1.} \tag{33}$$

If $L = 0$, then $y_2 = y_1$. For $L > 0$ and $t_0 \le t \le t_0 + \alpha$,

$$\begin{aligned}\sum_{n=1}^{\infty} |y_n(t) - y_{n-1}(t)| &\le \sum_{n=1}^{\infty} \frac{ML^{n-1}(t-t_0)^n}{n!} \\ &= \frac{M}{L} \sum_{n=1}^{\infty} \frac{[L(t-t_0)]^n}{n!} \\ &= \frac{M}{L}\left(e^{L(t-t_0)} - 1\right) \\ &\le \boxed{\frac{M}{L}\left(e^{\alpha L} - 1\right)} < \infty.\end{aligned}$$

5.3 Is Picard iterates right

Till now, we've proved that $\{y_n\}$ converges. However, we still don't know whether $\lim_{n\to\infty} y_n(t)$ is the answer of (29).
Consider $\lim_{n\to\infty} y_n(t) = y$, and we take limit of both sides of:

$$y_{n+1}(t) = y_0 + \int_{t_0}^{t} f(s, y_n(s))\, ds$$

Then we get

$$y = y_0 + \lim_{n\to\infty} \int_{t_0}^{t} f(s, y_n(s))\, ds \tag{34}$$

Then what we need to do is to justify the right hand side equals to $\int_{t_0}^{t} f(s, y(s))$.
One could also understand that the limit could *pass* the integral, as:

$$\lim_{n\to\infty} \int_{t_0}^{t} f(s, y_n(s))\, ds = \int_{t_0}^{t} \left(\lim_{n\to\infty} f(s, y_n(s))\right) ds = \int_{t_0}^{t} f(s, y(s))\, ds$$

We still need to justify this step. For $m > n$, by (33),

$$\begin{aligned}|y_m(t) - y_n(t)| &\le \sum_{k=n+1}^{m} |y_k(t) - y_{k-1}(t)| \\ &\le \sum_{k=n+1}^{m} \frac{ML^{k-1}(t-t_0)^k}{k!} \\ &\le \sum_{k=n+1}^{m} \frac{ML^{k-1}\alpha^k}{k!}.\end{aligned}$$

Let $m \to \infty$. Then

$$\sup_{t_0 \le t \le t_0+\alpha} |y(t) - y_n(t)| \le \sum_{k=n+1}^{\infty} \frac{ML^{k-1}\alpha^k}{k!} \longrightarrow 0. \tag{35}$$

Thus,

$$y_n \longrightarrow y \quad \text{uniformly on } [t_0, t_0+\alpha].$$

By the mean value theorem,

$$\begin{aligned}|f(s, y_n(s)) - f(s, y(s))| &\le L|y_n(s) - y(s)| \\ &\le L \sup_{t_0 \le r \le t_0+\alpha} |y_n(r) - y(r)|.\end{aligned}$$

Therefore,

$$\begin{aligned}&\left|\int_{t_0}^{t} f(s, y_n(s))\, ds - \int_{t_0}^{t} f(s, y(s))\, ds\right| \\ &\le \int_{t_0}^{t} |f(s, y_n(s)) - f(s, y(s))|\, ds \\ &\le \alpha L \sup_{t_0 \le r \le t_0+\alpha} |y_n(r) - y(r)| \\ &\longrightarrow 0.\end{aligned}$$

So the limit can pass through the integral:

$$\lim_{n\to\infty} \int_{t_0}^{t} f(s, y_n(s))\, ds = \int_{t_0}^{t} f(s, y(s))\, ds.$$

Taking the limit in the Picard iteration gives

$$\boxed{y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))\, ds.} \tag{36}$$

We also need to show that $y(t)$ is continuous. First,

$$y_0(t) = y_0 \quad \Longrightarrow \quad y_0 \text{ is continuous.}$$

If y_n is continuous, then

$$s \longmapsto f(s, y_n(s)) \text{ is continuous}$$

and

$$y_{n+1}(t) = y_0 + \int_{t_0}^{t} f(s, y_n(s))\, ds \quad \Longrightarrow \quad y_{n+1} \text{ is continuous.}$$

Thus, every y_n is continuous. Fix $t^* \in [t_0, t_0 + \alpha]$. For every $\varepsilon > 0$, the uniform convergence in (35) gives an N such that

$$\sup_{t_0 \le t \le t_0 + \alpha} |y(t) - y_N(t)| < \frac{\varepsilon}{3}.$$

Since y_N is continuous, there exists $\delta > 0$ such that

$$|t - t^*| < \delta \quad \Longrightarrow \quad |y_N(t) - y_N(t^*)| < \frac{\varepsilon}{3}.$$

Therefore,

$$\begin{aligned} |y(t) - y(t^*)| &\le |y(t) - y_N(t)| + |y_N(t) - y_N(t^*)| + |y_N(t^*) - y(t^*)| \\ &< \varepsilon. \end{aligned}$$

Hence,

$$\boxed{y \in C([t_0, t_0 + \alpha]).} \tag{37}$$

Finally,

$$y(t_0) = y_0, \qquad \frac{dy}{dt} = f(t, y(t)).$$

Thus, $y(t)$ is a solution of (29).

5.4 Uniqueness

Theorem 1.5.2. *Let f and $\partial f/\partial y$ be continuous in the rectangle*

$$R: \quad t_0 \le t \le t_0 + a, \quad |y - y_0| \le b.$$

Compute

$$M = \max_{(t,y)\in R} |f(t, y)|, \quad \text{and set } \alpha = \min\left(a, \frac{b}{M}\right).$$

Then, the initial-value problem

$$y' = f(t, y), \quad y(t_0) = y_0 \tag{38}$$

has a unique solution $y(t)$ on the interval $t_0 \le t \le t_0 + \alpha$. In other words, if $y(t)$ and $z(t)$ are two solutions of (38), then $y(t)$ must equal $z(t)$ for $t_0 \le t \le t_0 + \alpha$.

Proof. Theorem 2 guarantees the existence of at least one solution $y(t)$ of (38).

Suppose that $z(t)$ is a second solution of (38). Then,

$$y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))\, ds \quad \text{and} \quad z(t) = y_0 + \int_{t_0}^{t} f(s, z(s))\, ds.$$

Subtracting these two equations gives

$$\begin{aligned} |y(t) - z(t)| &= \left| \int_{t_0}^{t} [f(s, y(s)) - f(s, z(s))]\ ds \right| \\ &\leq \int_{t_0}^{t} |f(s, y(s)) - f(s, z(s))|\, ds \\ &\leq L \int_{t_0}^{t} |y(s) - z(s)|\, ds, \end{aligned}$$

where L is the maximum value of $|\partial f/\partial y|$ for (t, y) in R. As Lemma 5.3 below shows, this inequality implies that $y(t) = z(t)$. Hence, the initial-value problem (38) has a unique solution $y(t)$. □

Lemma 5.3. *Let $w(t)$ be a nonnegative function, with*

$$w(t) \leq L \int_{t_0}^{t} w(s)\, ds.$$

Then, $w(t)$ is identically zero.

Proof. Set

$$U(t) = \int_{t_0}^{t} w(s)\, ds.$$

Then

$$U'(t) = w(t) \leq L \int_{t_0}^{t} w(s)\, ds = LU(t).$$

Therefore,

$$\begin{aligned} \frac{d}{dt}\left(e^{-L(t-t_0)}U(t)\right) &= e^{-L(t-t_0)}\left(U'(t) - LU(t)\right) \\ &\leq 0. \end{aligned}$$

Thus,

$$0 \leq e^{-L(t-t_0)}U(t) \leq U(t_0) = 0,$$

so $U(t) = 0$. Finally,

$$0 \leq w(t) \leq L \int_{t_0}^{t} w(s)\, ds = LU(t) = 0.$$

Hence, $\boxed{w(t) = 0}$ for $t_0 \leq t \leq t_0 + \alpha$. □

5.5 Proof of (32)

First, consider a sequence $\{a_n\}_{n\geq 1} \subset \mathbb{R}$ such that

$$\sum_{n=1}^{\infty} |a_n| < \infty.$$

Define

$$S_n = \sum_{k=1}^{n} a_k, \qquad A_n = \sum_{k=1}^{n} |a_k|.$$

Since $\{A_n\}$ converges, it is a Cauchy sequence. Thus,

$$\forall \varepsilon > 0, \qquad \exists N \in \mathbb{N}, \qquad m > n \geq N$$

$$\Downarrow$$

$$|A_m - A_n| = \sum_{k=n+1}^{m} |a_k| < \varepsilon.$$

Then

$$\begin{aligned} |S_m - S_n| &= \left| \sum_{k=n+1}^{m} a_k \right| \\ &\leq \sum_{k=n+1}^{m} |a_k| \\ &< \varepsilon. \end{aligned}$$

Therefore,

$$\{S_n\} \text{ is Cauchy} \overset{\mathbb{R} \text{ is complete}}{\Longrightarrow} S_n \longrightarrow S \in \mathbb{R},$$

and hence

$$\boxed{\sum_{n=1}^{\infty} |a_n| < \infty \quad \Longrightarrow \quad \sum_{n=1}^{\infty} a_n \text{ converges.}} \tag{39}$$

Chapter 2

Second Order Linear Differential Equations

1 Introduction

We've considered first order differential equations. Now we try to understand second order.
Consider

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = f(t, y, \frac{\mathrm{d}y}{\mathrm{d}t}), \quad y(t_0) = y_0, y'(t_0) = y'_0 \tag{1}$$

Think about how we define linear first order equation before. Now we define second order linear differential equation.

Definition 2.1.1 (Second order linear differential equation)**.**
The equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = g(t) \tag{2}$$

is called second order linear differential equation.

Sadly, almost all second order differential equations are extremely difficult to solve. Till now we can only solve (2)

Consider an easier case of (2) when $g(t) = 0$. Like what we did in forst order, we can also define second order linear homogeneous equation.

Definition 2.1.2 (Second order linear homogeneous equation)**.**
The equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = 0 \tag{3}$$

is called second order linear homogeneous equation.

However, it could be very difficult even in solving second order linear homogeneous equation. We'll first consider whether it actually has a solution (or solutions) and its properties.

2 Algebraic properties of solutions

Consider

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = 0 \tag{4}$$

Or better, consider initial value problem:

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = 0, \quad y_0 = y(t_0), y_0' = y(t_0) \tag{5}$$

Then we consider an operator $L[y](t)$:

$$L[y](t) = \frac{\mathrm{d}^2y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y \tag{6}$$

One can understand $L[y](t)$ as 'function on functions'. We input a function $y(t)$, then output a function $L[y](t)$. With the help of operator, we could analysis the solutions of (4) easier.

Property 1 $L[cy] = cL[y]$
Property 2 $L[y_1 + y_2] = L[y_1] + L[y_2]$

Definition 2.2.1 (Linear operator)**.**
An operator L which satisfies properties 1 and 2 is called a linear operator. All other operators are non-linear.

Now we give a theorem which will be proved very later:

Theorem 2.2.1 (Existence–uniqueness Theorem)**.** *Let the functions $p(t)$ and $q(t)$ be continuous in the open interval $\alpha < t < \beta$. Then, there exists one, and only one function $y(t)$ satisfying the differential equation (4) on the entire interval $\alpha < t < \beta$, and the prescribed initial conditions*

$$y(t_0) = y_0, \qquad y'(t_0) = y_0'.$$

In particular, any solution $y = y(t)$ of (3) which satisfies $y(t_0) = 0$ and $y'(t_0) = 0$ at some time $t = t_0$ must be identically zero.

Thus the solution of (4) could be written in the form of (6) as

$$L[y](t) = 0 \tag{7}$$

Note the right hand side is also a function instead a scalar.

For any solution $y(t)$ of (7), $cy(t)$ is also the solution. For any solutions $y_1(t)$, $y_2(t)$ of (7), $c_1y_1(t) + c_2y_2(t)$ is also the solution.

Assume we already obtain two solutions y_1 and y_2 of (5), and $y(t)$ is any solution of (5). We must find constants c_1 and c_2 such that $y(t) = c_1y_1(t) + c_2y_2(t)$. The constants c_1 and c_2, if they exist, must satisfy the two equations

$$c_1 y_1(t_0) + c_2 y_2(t_0) = y_0,$$

$$c_1 y_1'(t_0) + c_2 y_2'(t_0) = y_0'.$$

Then it gives

$$c_1 \big[y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)\big] = y_0 y_2'(t_0) - y_0' y_2(t_0),$$

$$c_2 \big[y_1'(t_0)y_2(t_0) - y_1(t_0)y_2'(t_0)\big] = y_0 y_1'(t_0) - y_0' y_1(t_0).$$

Hence,

$$c_1 = \frac{y_0 y_2'(t_0) - y_0' y_2(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)},$$

$$c_2 = \frac{y_0' y_1(t_0) - y_0 y_1'(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)}$$

if $y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0) \neq 0$.
Then, we come to **Theorem 2.2.2**.

Theorem 2.2.2. *Let $y_1(t)$ and $y_2(t)$ be two solutions of (4) on the interval $\alpha < t < \beta$, with*

$$y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

unequal to zero in this interval. Then,

$$y(t) = c_1 y_1(t) + c_2 y_2(t)$$

is the general solution of (4).

Above we've already showed the idea of the proof. The entire proof will be in 2.1.

Definition 2.2.2 (Wronskian)**.**
The quantity

$$y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

is called the Wronskian *of y_1 and y_2, and is denoted by*

$$W(t) = W[y_1, y_2](t).$$

One could consider a matrix form:

$$\begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} c_1 y_1 + c_2 y_2 \\ c_1 y_1' + c_2 y_2' \end{bmatrix} = \begin{bmatrix} y \\ y' \end{bmatrix} \tag{8}$$

Then we come to Theorem 2.2.3 which shows some property of Wronskian, and we would prove Theorem 2.2.3 with the aid of Lemma 2.4.

Theorem 2.2.3. *Let $p(t)$ and $q(t)$ be continuous in the interval $\alpha < t < \beta$, and let $y_1(t)$ and $y_2(t)$ be two solutions of (3). Then, $W[y_1, y_2](t)$ is either identically zero, or is never zero, on the interval $\alpha < t < \beta$.*

Lemma 2.4. *Let $y_1(t)$ and $y_2(t)$ be two solutions of the linear differential equation*

$$y'' + p(t)y' + q(t)y = 0.$$

Then, their Wronskian

$$W(t) = W[y_1, y_2](t) = y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

satisfies the first-order differential equation

$$W' + p(t)W = 0.$$

Entire proof will be in 2.2.

Easy to see, if $y_1 = cy_2$, then $W[y_1, y_2](t) = 0$. Also, it could be proved that if $W[y_1, y_2](t) = 0$, then $y_1 = cy_2$.

Theorem 2.2.5. *Let $y_1(t)$ and $y_2(t)$ be two solutions of (3) on the interval $\alpha < t < \beta$, and suppose that $W[y_1, y_2](t_0) = 0$ for some t_0 in this interval. Then, one of these solutions is a constant multiple of the other.*

Entire proof will be in 2.3.

Definition 2.2.3 (Linear dependent)**.**
The functions $y_1(t)$ and $y_2(t)$ are said to be linearly dependent *on an interval I if one of these functions is a constant multiple of the other on I. The functions $y_1(t)$ and $y_2(t)$ are said to be* linearly independent *on an interval I if they are not linearly dependent on I.*

Lemma 2.6 (to Theorem 2.2.5)**.** *Two solutions $y_1(t)$ and $y_2(t)$ of (3) are linearly independent on the interval $\alpha < t < \beta$ if, and only if, their Wronskian is not equal to zero on this interval. Thus, two solutions $y_1(t)$ and $y_2(t)$ form a fundamental set of solutions of (3) on the interval $\alpha < t < \beta$ if, and only if, they are linearly independent on this interval.*

2.1 Proof of Theorem 2.2.2

Theorem 2.2.2. *Let $y_1(t)$ and $y_2(t)$ be two solutions of (4) on the interval $\alpha < t < \beta$, with*

$$y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

unequal to zero in this interval. Then,

$$y(t) = c_1y_1(t) + c_2y_2(t)$$

is the general solution of (4).

Proof. Let $y(t)$ be any solution of (4). We must find constants c_1 and c_2 such that $y(t) = c_1y_1(t) + c_2y_2(t)$. To this end, pick a time t_0 in the interval (α, β) and let y_0 and y_0' denote the values of y and y' at $t = t_0$. The constants c_1 and c_2, if they exist, must satisfy the two equations

$$c_1y_1(t_0) + c_2y_2(t_0) = y_0,$$

$$c_1y_1'(t_0) + c_2y_2'(t_0) = y_0'.$$

Multiplying the first equation by $y_2'(t_0)$, the second equation by $y_2(t_0)$, and subtracting gives

$$c_1\left[y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)\right] = y_0y_2'(t_0) - y_0'y_2(t_0).$$

Similarly, multiplying the first equation by $y_1'(t_0)$, the second equation by $y_1(t_0)$, and subtracting gives

$$c_2\left[y_1'(t_0)y_2(t_0) - y_1(t_0)y_2'(t_0)\right] = y_0y_1'(t_0) - y_0'y_1(t_0).$$

Hence,

$$c_1 = \frac{y_0y_2'(t_0) - y_0'y_2(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)}$$

and

$$c_2 = \frac{y_0'y_1(t_0) - y_0y_1'(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)}$$

if $y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0) \neq 0$.
Now, let

$$\phi(t) = c_1y_1(t) + c_2y_2(t)$$

for this choice of constants c_1, c_2. We know that $\phi(t)$ satisfies (4), since it is a linear combination of solutions of (4). Moreover, by construction, $\phi(t_0) = y_0$ and $\phi'(t_0) = y_0'$. Thus, $y(t)$ and $\phi(t)$ satisfy the same second-order linear homogeneous equation and the same initial conditions. Therefore, by the uniqueness part of Theorem 2.2.1, $y(t)$ must be identically equal to $\phi(t)$; that is,

$$y(t) = c_1y_1(t) + c_2y_2(t), \quad \alpha < t < \beta.$$

Theorem 2 is an extremely useful theorem since it reduces the problem of finding all solutions of (4), of which there are infinitely many, to the much simpler problem of finding just two solutions $y_1(t), y_2(t)$. The only condition imposed on the solutions

$y_1(t)$ and $y_2(t)$ is that the quantity

$$y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

be unequal to zero for $\alpha < t < \beta$. When this is the case, we say that $y_1(t)$ and $y_2(t)$ are a *fundamental set of solutions* of (4), since all other solutions of (4) can be obtained by taking linear combinations of $y_1(t)$ and $y_2(t)$. □

2.2 Proof of Theorem 2.2.3 and Lemma 2.4

Lemma 2.4. *Let $y_1(t)$ and $y_2(t)$ be two solutions of the linear differential equation*

$$y'' + p(t)y' + q(t)y = 0.$$

Then, their Wronskian

$$W(t) = W[y_1, y_2](t) = y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

satisfies the first-order differential equation

$$W' + p(t)W = 0.$$

Proof. Observe that

$$W'(t) = \frac{d}{dt}(y_1y_2' - y_1'y_2) = y_1y_2'' + y_1'y_2' - y_1'y_2' - y_1''y_2 = y_1y_2'' - y_1''y_2.$$

Since y_1 and y_2 are solutions of $y'' + p(t)y' + q(t)y = 0$, we know that

$$y_2'' = -p(t)y_2' - q(t)y_2$$

and

$$y_1'' = -p(t)y_1' - q(t)y_1.$$

Hence,

$$W'(t) = y_1\left[-p(t)y_2' - q(t)y_2\right] - y_2\left[-p(t)y_1' - q(t)y_1\right] = -p(t)\left[y_1y_2' - y_1'y_2\right] = -p(t)W(t).$$

□

Theorem 2.2.3. *Let $p(t)$ and $q(t)$ be continuous in the interval $\alpha < t < \beta$, and let $y_1(t)$ and $y_2(t)$ be two solutions of (3). Then, $W[y_1, y_2](t)$ is either identically zero, or is never zero, on the interval $\alpha < t < \beta$.*

Proof of Theorem 3. Pick any t_0 in the interval (α, β). From Lemma 1,

$$W[y_1, y_2](t) = W[y_1, y_2](t_0)\exp\left(-\int_{t_0}^{t} p(s)\,ds\right).$$

Now, $\exp\left(-\int_{t_0}^{t} p(s)\,ds\right)$ is unequal to zero for $\alpha < t < \beta$. Therefore, $W[y_1, y_2](t)$ is either identically zero, or is never zero. □

2.3 Proof of Theorem 2.2.5

Theorem 2.2.5. *Let $y_1(t)$ and $y_2(t)$ be two solutions of (3) on the interval $\alpha < t < \beta$, and suppose that $W[y_1, y_2](t_0) = 0$ for some t_0 in this interval. Then, one of these solutions is a constant multiple of the other.*

Here we provide two method to prove:

Proof #1. Suppose that $W[y_1, y_2](t_0) = 0$. Then, recall the matrix form, the equations

$$c_1 y_1(t_0) + c_2 y_2(t_0) = 0,$$
$$c_1 y_1'(t_0) + c_2 y_2'(t_0) = 0$$

have a nontrivial solution c_1, c_2; that is, a solution c_1, c_2 with $|c_1| + |c_2| \neq 0$. Let $y(t) = c_1 y_1(t) + c_2 y_2(t)$, for this choice of constants c_1, c_2. We know that $y(t)$ is a solution of (3), since it is a linear combination of $y_1(t)$ and $y_2(t)$. Moreover, by construction, $y(t_0) = 0$ and $y'(t_0) = 0$. Therefore, by Theorem 2.2.1, $y(t)$ is identically zero, so that

$$c_1 y_1(t) + c_2 y_2(t) = 0, \quad \alpha < t < \beta.$$

If $c_1 \neq 0$, then $y_1(t) = -(c_2/c_1) y_2(t)$, and if $c_2 \neq 0$, then $y_2(t) = -(c_1/c_2) y_1(t)$. In either case, one of these solutions is a constant multiple of the other. □ □

Proof #2. Suppose that $W[y_1, y_2](t_0) = 0$. Then, by Theorem 2.2.3, $W[y_1, y_2](t)$ is identically zero. Assume that $y_1(t) y_2(t) \neq 0$ for $\alpha < t < \beta$. Then, dividing both sides of the equation

$$y_1(t) y_2'(t) - y_1'(t) y_2(t) = 0$$

by $y_1(t) y_2(t)$ gives

$$\frac{y_2'(t)}{y_2(t)} - \frac{y_1'(t)}{y_1(t)} = 0.$$

This equation implies that $y_1(t) = c y_2(t)$ for some constant c.
Next, suppose that $y_1(t) y_2(t)$ is zero at some point $t = t^*$ in the interval $\alpha < t < \beta$. Without loss of generality, we may assume that $y_1(t^*) = 0$, since otherwise we can relabel y_1 and y_2. In this case it is simple to show that either $y_1(t) \equiv 0$, or

$$y_2(t) = \frac{y_2'(t^*)}{y_1'(t^*)} y_1(t).$$

This completes the proof of Theorem 2.2.5. □ □

3 Linear differential equations with constant coefficients

Consider

$$L[y](t) = a\frac{\mathrm{d}^2 y(t)}{\mathrm{d}t^2} + b\frac{\mathrm{d}y(t)}{\mathrm{d}t} + cy(t) = 0, \qquad a \neq 0, \tag{9}$$

where a, b, c are constants.
By Theorem 2.2.2, we know that we only need to find two linearly independent solutions of (9). However, the theorem does not tell us how to find them.
Easy to see, that y'', y', y should be the same type. One natural idea is to try:

$$y(t) = e^{rt}.$$

Then

$$\begin{aligned} L[e^{rt}] &= a\frac{d^2}{dt^2}e^{rt} + b\frac{d}{dt}e^{rt} + ce^{rt} \\ &= ar^2e^{rt} + bre^{rt} + ce^{rt} \\ &= (ar^2 + br + c)e^{rt}. \end{aligned}$$

Since $e^{rt} \neq 0$,

$$L[e^{rt}] = 0 \quad \Longleftrightarrow \quad ar^2 + br + c = 0.$$

Definition 2.3.1 (Characteristic equation)**.**
The equation

$$ar^2 + br + c = 0 \tag{10}$$

is called the characteristic equation of (9).

The roots are

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \qquad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}. \tag{11}$$

Thus, the form of the solution depends on

$$\Delta = b^2 - 4ac.$$

We consider the three cases

$$\Delta > 0, \qquad \Delta < 0, \qquad \Delta = 0.$$

3.1 Real and distinct roots

Suppose

$$\Delta = b^2 - 4ac > 0.$$

Then r_1, r_2 in (11) are real and $r_1 \neq r_2$. Therefore,

$$y_1(t) = e^{r_1 t}, \qquad y_2(t) = e^{r_2 t}$$

are two solutions of (9).
Very easy to see, y_1 is not a constant multiple of y_2.
Thus, y_1, y_2 form a fundamental set of solutions. By Theorem 2.2.2,

$$\boxed{y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}.} \tag{12}$$

3.2 Complex roots

There are two solutions:

$$r_1 = \frac{-b + i\sqrt{-b^2 + 4ac}}{2a}, \qquad r_2 = \frac{-b - i\sqrt{-b^2 + 4ac}}{2a}. \tag{13}$$

We use two notions:

$$\alpha = -\frac{b}{2a}, \quad \beta = \frac{\sqrt{4ac - b^2}}{2a}$$

However, there are two problem remaining to be solved:
Problem 1 e^{rt} has not been well-defined yet for complex r.
Problem 2 Even e^{rt} is well-defined, we still need to find the real-value solution.
First, we try to solve **Problem 1**.

Lemma 3.1. *Suppose that $a, b, c \in \mathbb{R}$ and*

$$y(t) = u(t) + iv(t)$$

is a complex-valued solution of (9). Then $u(t)$ and $v(t)$ are real-valued solutions of (9).

Proof. Since $L[y] = 0$,

$$\begin{aligned} 0 &= L[u + iv] \\ &= L[u] + iL[v] \\ &= \big(au'' + bu' + cu\big) + i\big(av'' + bv' + cv\big). \end{aligned}$$

Thus,

$$au'' + bu' + cu = 0, \qquad av'' + bv' + cv = 0.$$

Hence, both u and v solve (9). □

Then, we try to solve **Problem 2**. Now we write $e^{(\alpha+i\beta)t}$ explicitly. From the power series,

$$\begin{aligned} e^{i\beta t} &= 1 + i\beta t + \frac{(i\beta t)^2}{2!} + \frac{(i\beta t)^3}{3!} + \cdots \\ &= \left(1 - \frac{\beta^2 t^2}{2!} + \frac{\beta^4 t^4}{4!} - \cdots\right) i \left(\beta t - \frac{\beta^3 t^3}{3!} + \frac{\beta^5 t^5}{5!} - \cdots\right) \\ &= \cos(\beta t) + i\sin(\beta t). \end{aligned}$$

Therefore,

$$e^{(\alpha+i\beta)t} = e^{\alpha t}\big(\cos(\beta t) + i\sin(\beta t)\big). \tag{14}$$

Also,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}e^{(\alpha+i\beta)t} &= e^{\alpha t}\left[\alpha\cos(\beta t) - \beta\sin(\beta t) + i\big(\alpha\sin(\beta t) + \beta\cos(\beta t)\big)\right] \\ &= (\alpha+i\beta)e^{(\alpha+i\beta)t}.\end{aligned}$$

Since $r_1 = \alpha + i\beta$ satisfies the characteristic equation, $e^{r_1 t}$ is a complex-valued solution of (9). By Lemma 3.1, its real and imaginary parts

$$y_1(t) = e^{\alpha t}\cos(\beta t), \qquad y_2(t) = e^{\alpha t}\sin(\beta t)$$

are real-valued solutions.
Also,

$$\begin{aligned}W[y_1,y_2](t) &= \begin{vmatrix} e^{\alpha t}\cos(\beta t) & e^{\alpha t}\sin(\beta t) \\ e^{\alpha t}\big(\alpha\cos(\beta t) - \beta\sin(\beta t)\big) & e^{\alpha t}\big(\alpha\sin(\beta t) + \beta\cos(\beta t)\big)\end{vmatrix} \\ &= \beta e^{2\alpha t} \\ &\neq 0.\end{aligned}$$

Thus,

$$\boxed{y(t) = e^{\alpha t}\left[c_1\cos(\beta t) + c_2\sin(\beta t)\right].} \tag{15}$$

3.3 Equal roots

Suppose

$$\Delta = b^2 - 4ac = 0.$$

Then the characteristic equation has only one root

$$r = -\frac{b}{2a},$$

so the guess $y = e^{rt}$ only gives one solution

$$y_1(t) = e^{-bt/(2a)}.$$

We still need a second solution which is linearly independent of y_1.
More generally, suppose we already know one solution $y_1(t)$ of

$$y'' + p(t)y' + q(t)y = 0. \tag{16}$$

We try to use y_1 to construct another solution. Set

$$y(t) = y_1(t)v(t).$$

Then

$$y' = y_1'v + y_1v', \qquad y'' = y_1''v + 2y_1'v' + y_1v''.$$

Substituting into (16) gives

$$\begin{aligned} 0 &= \big(y_1'' + py_1' + qy_1\big)v + \big(2y_1' + py_1\big)v' + y_1v'' \\ &= \big(2y_1' + py_1\big)v' + y_1v'', \end{aligned}$$

since y_1 is a solution.
Set

$$u = v'.$$

Then

$$y_1u' + (2y_1' + py_1)u = 0,$$

or

$$u' + \left(2\frac{y_1'}{y_1} + p\right)u = 0.$$

This is a first-order linear homogeneous equation. Thus,

$$\begin{aligned} u(t) &= \exp\left[-\int\left(2\frac{y_1'}{y_1} + p(t)\right)dt\right] \\ &= \frac{\exp\big(-\int p(t)\,dt\big)}{y_1^2(t)}. \end{aligned}$$

Since $u = v'$, we can take

$$v(t) = \int \frac{\exp\big(-\int p(t)\,dt\big)}{y_1^2(t)}\,dt.$$

Therefore, a second solution is

$$\boxed{y_2(t) = y_1(t)\int \frac{\exp\big(-\int p(t)\,dt\big)}{y_1^2(t)}\,dt.} \tag{17}$$

This method changes the problem from a second-order equation to a first-order equation, so it is called *reduction of order*.
Now return to (9). Dividing it by a, we obtain

$$y'' + \frac{b}{a}y' + \frac{c}{a}y = 0.$$

Thus,

$$p(t) = \frac{b}{a}, \qquad y_1(t) = e^{-bt/(2a)}.$$

The integrand in (17) becomes

$$\begin{aligned} \frac{\exp\big(-\int p(t)\,dt\big)}{y_1^2(t)} &= \frac{e^{-bt/a}}{\big(e^{-bt/(2a)}\big)^2} \\ &= 1. \end{aligned}$$

Therefore,

$$v(t) = \int 1\,dt = t, \qquad y_2(t) = te^{-bt/(2a)}.$$

Their Wronskian is

$$W[y_1, y_2](t) = e^{-bt/a} \neq 0.$$

Hence, the general solution for equal roots is

$$\boxed{y(t) = c_1 e^{-bt/(2a)} + c_2 t e^{-bt/(2a)} = (c_1 + c_2 t)\, e^{-bt/(2a)}.} \tag{18}$$

For example, consider

$$y'' + 4y' + 4y = 0, \qquad y(0) = 1, \quad y'(0) = 3.$$

The characteristic equation is

$$r^2 + 4r + 4 = (r+2)^2 = 0.$$

Thus,

$$y(t) = c_1 e^{-2t} + c_2 t e^{-2t}.$$

The initial conditions give

$$c_1 = 1, \qquad -2c_1 + c_2 = 3.$$

So

$$c_1 = 1, \qquad c_2 = 5,$$

and

$$\boxed{y(t) = (1+5t)e^{-2t}.}$$